Parametric Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Eliminating the parameter**

Eliminate the parameter from $x(t) = t$, $y(t) = t^2 - 1$ to find the Cartesian equation, and sketch it.

Since $x = t$, substitute directly: $y = x^2 - 1$, an upward parabola shifted down 1 unit.

2 **[Foundational]** • **Target: Parametric equations from real motion**

A Ferris wheel of radius 10 m completes one revolution every 40 s, centered 12 m above the ground. Write parametric equations for a rider's height above the ground as a function of time t (seconds), assuming the rider starts at the bottom.

Angular speed $= \frac{2\pi}{40} = \frac{\pi}{20}$ rad/s. Starting at the bottom means height $= 12 - 10\cos\left(\frac{\pi}{20}t\right)$ (using $x(t) = t$ for time, $y(t) = 12 - 10\cos\left(\frac{\pi}{20}t\right)$ for height).

3 **[Foundational]** • **Target: Derivatives of parametric equations**

For $x(t) = t^2$, $y(t) = t^3$, find $\frac{dy}{dx}$ in terms of t.

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2}{2t} = \frac{3t}{2}$.

4 **[Foundational]** • **Target: Derivatives of parametric equations**

For $x(t) = \cos t$, $y(t) = \sin t$, find $\frac{dy}{dx}$ and evaluate it at $t = \frac{\pi}{4}$.

$\frac{dy}{dx} = \frac{\cos t}{-\sin t} = -\cot t$. At $t = \frac{\pi}{4}$: $-\cot\left(\frac{\pi}{4}\right) = -1$.

5 **[Foundational]** • **Target: Second derivatives of parametric equations**

For $x(t) = t^2$, $y(t) = t^3$, find $\frac{d^2y}{dx^2}$ using $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$.

$\frac{dy}{dx} = \frac{3t}{2}$; $\frac{d}{dt}\left(\frac{3t}{2}\right) = \frac{3}{2}$. $\frac{d^2y}{dx^2} = \frac{3/2}{2t} = \frac{3}{4t}$.

6 **[Foundational]** • **Target: Tangent lines to parametric curves**

Find the equation of the tangent line to the curve $x(t) = t^2 - 1$, $y(t) = t^3 - t$ at $t = 2$.

At $t = 2$: $x = 3$, $y = 6$. $\frac{dx}{dt} = 2t = 4$; $\frac{dy}{dt} = 3t^2 - 1 = 11$. $\frac{dy}{dx} = \frac{11}{4}$. Tangent: $y - 6 = \frac{11}{4}(x - 3)$.

7 **[Foundational]** • **Target: Arc length of parametric curves**

State the arc length formula for a parametric curve $x(t)$, $y(t)$ on $[a, b]$.

$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Arc length of parametric curves**

Find the arc length of $x(t) = 3t$, $y(t) = 4t$ for $t \in [0, 2]$.

$\frac{dx}{dt} = 3$, $\frac{dy}{dt} = 4$. $L = \int_0^2 \sqrt{9 + 16}\, dt = \int_0^2 5\, dt = 10$.

9 **[Intermediate]** • **Target: Arc length of parametric curves**

Set up (but do not evaluate) the arc length integral for $x(t) = t^2$, $y(t) = t^3$ on $[0, 1]$.

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 3t^2$. $L = \int_0^1 \sqrt{(2t)^2 + (3t^2)^2}\, dt = \int_0^1 \sqrt{4t^2 + 9t^4}\, dt$.

10 **[Intermediate]** • **Target: Speed as a vector quantity**

A particle moves with position $x(t) = t^2$, $y(t) = 2t$. Find its speed at $t = 1$.

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 2$. At $t = 1$: $\frac{dx}{dt} = 2$, $\frac{dy}{dt} = 2$. Speed $= \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$.

11 **[Intermediate]** • **Target: Speed as a vector quantity**

A particle moves with position $x(t) = 3\cos t$, $y(t) = 3\sin t$. Find its speed at any time t, and interpret the result.

$\frac{dx}{dt} = -3\sin t$, $\frac{dy}{dt} = 3\cos t$. Speed $= \sqrt{9\sin^2 t + 9\cos^2 t} = \sqrt{9} = 3$ (constant). The particle traces a circle of radius 3 at constant speed.

12 **[Intermediate]** • **Target: Eliminating the parameter**

Eliminate the parameter from $x(t) = 2t + 1$, $y(t) = 4t - 3$ to find the Cartesian equation.

From $x = 2t + 1$: $t = \frac{x-1}{2}$. Substitute: $y = 4\left(\frac{x-1}{2}\right) - 3 = 2(x - 1) - 3 = 2x - 5$. A line with slope 2.

13 **[Intermediate]** • **Target: Eliminating the parameter**

Explain why eliminating the parameter from $x(t) = \cos t$, $y(t) = \sin t$ using the identity $\sin^2 t + \cos^2 t = 1$ gives a relation that is NOT a function $y = f(x)$.

Substituting gives $x^2 + y^2 = 1$, the equation of a unit circle. For most x-values in $(-1, 1)$, there are two corresponding y-values ($\pm\sqrt{1 - x^2}$), so this relation fails the vertical line test and is not a function of x.

14 **[Intermediate]** • **Target: Concavity of parametric curves**

For $x(t) = t^2$, $y(t) = t^3$ (with $\frac{d^2y}{dx^2} = \frac{3}{4t}$ found earlier), determine the concavity of the curve for $t > 0$ and for $t < 0$.

For $t > 0$: $\frac{d^2y}{dx^2} = \frac{3}{4t} > 0$, so the curve is concave up. For $t < 0$: $\frac{3}{4t} < 0$, so the curve is concave down.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Horizontal and vertical tangents** • **Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests**

For $x(t) = t^2 - 4$, $y(t) = t^3 - 3t$, find the value(s) of t where the tangent line is horizontal (where $\frac{dy}{dt} = 0$, provided $\frac{dx}{dt} \neq 0$).

$\frac{dy}{dt} = 3t^2 - 3 = 0 \Rightarrow t = \pm 1$. Check $\frac{dx}{dt} = 2t$: at $t = \pm 1$, $\frac{dx}{dt} = \pm 2 \neq 0$, so both give valid horizontal tangents.

16 **[Challenging]** • **Target: Horizontal and vertical tangents** • **Cross-Lesson: 6.01 Power Series and Interval of Convergence**

For the same curve, find the value(s) of t where the tangent line is vertical (where $\frac{dx}{dt} = 0$, provided $\frac{dy}{dt} \neq 0$).

$\frac{dx}{dt} = 2t = 0 \Rightarrow t = 0$. Check $\frac{dy}{dt} = 3(0)^2 - 3 = -3 \neq 0$, so $t = 0$ gives a valid vertical tangent, at the point $x(0) = -4$, $y(0) = 0$.

17 **[Challenging]** • **Target: Motion along a parametric path (particle interpretation)** • **Cross-Lesson: 6.02 Taylor and Maclaurin Series**

A particle's position is $x(t) = t^2$, $y(t) = t^3 - 3t$ for $t \in [-2, 2]$. Find all times when the particle is moving straight up or down (vertical velocity component only, i.e. $\frac{dx}{dt} = 0$).

$\frac{dx}{dt} = 2t = 0 \Rightarrow t = 0$. At $t = 0$: position is $(0, 0)$, and the particle is momentarily moving purely vertically (since horizontal velocity is 0 there).

18 **[Challenging]** • **Target: Motion along a parametric path (particle interpretation)** • **Cross-Lesson: 1.01 Integration by Parts**

For the particle above, find the speed at $t = 2$.

$\frac{dx}{dt} = 2t = 4$; $\frac{dy}{dt} = 3t^2 - 3 = 9$. Speed $= \sqrt{4^2 + 9^2} = \sqrt{16 + 81} = \sqrt{97} \approx 9.85$.

19 **[Challenging]** • **Target: Arc length of a full circle (verifying the circumference formula)** • **Cross-Lesson: 1.02 Partial Fractions and Improper Integrals**

Use the parametric arc length formula to verify that the circumference of a circle $x(t) = r\cos t$, $y(t) = r\sin t$ for $t \in [0, 2\pi]$ is $2\pi r$.

$\frac{dx}{dt} = -r\sin t$, $\frac{dy}{dt} = r\cos t$. $L = \int_0^{2\pi} \sqrt{r^2\sin^2 t + r^2\cos^2 t}\, dt = \int_0^{2\pi} r\, dt = r(2\pi) = 2\pi r$, matching the known circumference formula.

20 **[Challenging]** • **Target: Arc length of a full circle (verifying the circumference formula)** • **Cross-Lesson: 2.01 Euler's Method and Logistic Growth**

Find the arc length of the parametric curve $x(t) = e^t\cos t$, $y(t) = e^t\sin t$ is a spiral; set up (but do not evaluate) its arc length integral on $[0, \pi]$.

$\frac{dx}{dt} = e^t\cos t - e^t\sin t$, $\frac{dy}{dt} = e^t\sin t + e^t\cos t$. After simplification, $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 2e^{2t}$. So $L = \int_0^{\pi} \sqrt{2e^{2t}}\, dt = \int_0^{\pi} e^t\sqrt{2}\, dt$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Derivatives of parametric equations

For $x(t) = 6t^2$, $y(t) = -2t^3$, find $\frac{dy}{dx}$ at $t = 3$.

a. $-\frac{3}{2}$ ✓ **b.** -6

c. $\frac{3}{2}$ **d.** $-\frac{1}{2}$

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-6t^2}{12t} = \frac{-6}{12}t$. At $t = 3$: $\frac{-6}{12}(3) = -\frac{3}{2}$.

22 [Foundational] • Target: Second derivatives of parametric equations

For $x(t) = 2t^2$, $y(t) = -2t^3$, find $\frac{dy}{dx}$ at $t = 6$.

a. -4 **b.** 9

c. -9 ✓ **d.** -8

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-6t^2}{4t} = \frac{-6}{4}t$. At $t = 6$: $\frac{-6}{4}(6) = -9$.

23 [Foundational] • Target: Tangent lines to parametric curves

For $x(t) = 4t^2$, $y(t) = -2t^3$, find $\frac{dy}{dx}$ at $t = 3$.

a. $\frac{9}{4}$ **b.** -4

c. $-\frac{9}{4}$ ✓ **d.** $-\frac{5}{4}$

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-6t^2}{8t} = \frac{-6}{8}t$. At $t = 3$: $\frac{-6}{8}(3) = -\frac{9}{4}$.

24 [Foundational] • Target: Arc length of parametric curves

For $x(t) = 4t^2$, $y(t) = t^3$, find $\frac{dy}{dx}$ at $t = 4$.

a. $\frac{5}{2}$ **b.** $\frac{3}{2}$ ✓

c. $\frac{32}{3}$ **d.** $-\frac{3}{2}$

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2}{8t} = \frac{3}{8}t$. At $t = 4$: $\frac{3}{8}(4) = \frac{3}{2}$.

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25 [Foundational] • Target: Speed as a vector quantity

For $x(t) = 6t^2$, $y(t) = -3t^3$, find $\frac{dy}{dx}$ at $t = 1$.

a. $-\frac{4}{3}$

b. $-\frac{3}{4}$ ✓

c. $\frac{1}{4}$

d. $\frac{3}{4}$

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-9t^2}{12t} = \frac{-9}{12}t$. At $t = 1$: $\frac{-9}{12}(1) = -\frac{3}{4}$.

26 [Foundational] • Target: Eliminating the parameter

For $x(t) = t^2$, $y(t) = 5t^3$, find $\frac{dy}{dx}$ at $t = 4$.

a. $\frac{8}{15}$

b. -30

c. 31

d. 30 ✓

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{15t^2}{2t} = \frac{15}{2}t$. At $t = 4$: $\frac{15}{2}(4) = 30$.

27 [Foundational] • Target: Concavity of parametric curves

For $x(t) = 6t^2$, $y(t) = 2t^3$, find $\frac{dy}{dx}$ at $t = 5$.

a. $-\frac{5}{2}$

b. $\frac{7}{2}$

c. 10

d. $\frac{5}{2}$ ✓

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{6t^2}{12t} = \frac{6}{12}t$. At $t = 5$: $\frac{6}{12}(5) = \frac{5}{2}$.

28 [Foundational] • Target: Horizontal and vertical tangents

Eliminate the parameter from $x(t) = -2t - 3$, $y(t) = 7t - 9$ to find the Cartesian equation, then state its slope.

a. $-\frac{7}{2}$ ✓

b. $-\frac{5}{2}$

c. $-\frac{2}{7}$

d. $\frac{7}{2}$

From $x = -2t - 3$: $t = \frac{x+3}{-2}$. Substituting into $y = 7t - 9$ gives a line of slope $\frac{7}{-2} = -\frac{7}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Motion along a parametric path (particle interpretation)

Eliminate the parameter from $x(t) = 2t - 9$, $y(t) = -t + 4$ to find the Cartesian equation, then state its slope.

a. $-\frac{1}{2}$ ✓

b. -2

c. $\frac{1}{2}$

d. 0

From $x = 2t - 9$: $t = \frac{x+9}{2}$. Substituting into $y = -t + 4$ gives a line of slope $\frac{-1}{2} = -\frac{1}{2}$.

30 [Intermediate] • Target: Arc length of a full circle (verifying the circumference formula)

Eliminate the parameter from $x(t) = -5t - 4$, $y(t) = -4t - 7$ to find the Cartesian equation, then state its slope.

a. $-\frac{4}{5}$

b. $\frac{5}{4}$

c. $\frac{4}{5}$ ✓

d. $\frac{9}{5}$

From $x = -5t - 4$: $t = \frac{x+4}{-5}$. Substituting into $y = -4t - 7$ gives a line of slope $\frac{-4}{-5} = \frac{4}{5}$.

31 [Intermediate] • Target: Parametric equations from real motion

Eliminate the parameter from $x(t) = t + 1$, $y(t) = 2t + 5$ to find the Cartesian equation, then state its slope.

a. 3

b. -2

c. $\frac{1}{2}$

d. 2 ✓

From $x = t + 1$: $t = \frac{x-1}{1}$. Substituting into $y = 2t + 5$ gives a line of slope $\frac{2}{1} = 2$.

32 [Intermediate] • Target: First and Second Derivatives Parametrically

Eliminate the parameter from $x(t) = -4t + 3$, $y(t) = -t + 5$ to find the Cartesian equation, then state its slope.

a. $-\frac{1}{4}$

b. $\frac{1}{4}$ ✓

c. 4

d. $\frac{5}{4}$

From $x = -4t + 3$: $t = \frac{x-3}{-4}$. Substituting into $y = -t + 5$ gives a line of slope $\frac{-1}{-4} = \frac{1}{4}$.

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33 [Intermediate] • Target: Horizontal and Vertical Tangents

Eliminate the parameter from $x(t) = 4t + 1$, $y(t) = 9t + 7$ to find the Cartesian equation, then state its slope.

a. $-\frac{9}{4}$

b. $\frac{13}{4}$

c. $\frac{9}{4}$ ✓

d. $\frac{4}{9}$

From $x = 4t + 1$: $t = \frac{x-1}{4}$. Substituting into $y = 9t + 7$ gives a line of slope $\frac{9}{4} = \frac{9}{4}$.

34 [Intermediate] • Target: Arc Length of Parametric Curves

Eliminate the parameter from $x(t) = 3t - 1$, $y(t) = 2t + 2$ to find the Cartesian equation, then state its slope.

a. $-\frac{2}{3}$

b. $\frac{2}{3}$ ✓

c. $\frac{5}{3}$

d. $\frac{3}{2}$

From $x = 3t - 1$: $t = \frac{x+1}{3}$. Substituting into $y = 2t + 2$ gives a line of slope $\frac{2}{3} = \frac{2}{3}$.

35 [Intermediate] • Target: Speed and Total Distance Traveled

For $x(t) = 3t^2 + 5$, $y(t) = t^3 - 4t$, find $\frac{dy}{dx}$ at $t = 1$.

a. $\frac{1}{6}$

b. $\frac{5}{6}$

c. -6

d. $-\frac{1}{6}$ ✓

$\frac{dx}{dt} = 6t$, $\frac{dy}{dt} = 3t^2 - 4$. At $t = 1$: $\frac{dx}{dt} = 6$, $\frac{dy}{dt} = -1$. $\frac{dy}{dx} = \frac{-1}{6} = -\frac{1}{6}$.

36 [Intermediate] • Target: Derivatives of parametric equations

For $x(t) = 3t^2 - 5$, $y(t) = 5t^3 + 3t$, find $\frac{dy}{dx}$ at $t = 3$.

a. $\frac{26}{3}$

b. $\frac{3}{23}$

c. $-\frac{23}{3}$

d. $\frac{23}{3}$ ✓

$\frac{dx}{dt} = 6t$, $\frac{dy}{dt} = 15t^2 + 3$. At $t = 3$: $\frac{dx}{dt} = 18$, $\frac{dy}{dt} = 138$. $\frac{dy}{dx} = \frac{138}{18} = \frac{23}{3}$.

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37 [Intermediate] • Target: Second derivatives of parametric equations

For $x(t) = t^2$, $y(t) = t^3 - 9t^2 + 24t$, find the larger value of t where the tangent line is horizontal.

a. 2

b. 4 ✓

c. 6

d. 8

$\frac{dy}{dt} = 3t^2 - 18t + 24 = 3(t-(2))(t-(4)) = 0$ gives $t = 2$ and $t = 4$; the larger is 4.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Tangent lines to parametric curves • Cross-Lesson: 1.01 Integration by Parts

For $x(t) = t^2 - 3t$, $y(t) = t^3 + 3t^2 - 24t$, find the larger value of t where the tangent line is horizontal.

a. -4

b. -2

c. -8

d. 2 ✓

$\frac{dy}{dt} = 3t^2 + 6t - 24 = 3(t-(-4))(t-(2)) = 0$ gives $t = -4$ and $t = 2$; the larger is 2.

39 [Challenging] • Target: Arc length of parametric curves • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

A particle moves with position $x(t) = t^2$, $y(t) = 6t$. Find its speed at $t = 4$.

a. 12

b. 8

c. 6

d. 10 ✓

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 6$. At $t = 4$: $\frac{dx}{dt} = 8$. Speed $= \sqrt{8^2 + 6^2} = \sqrt{100} = 10$.

40 [Challenging] • Target: Speed as a vector quantity • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

A particle moves with position $x(t) = t^2$, $y(t) = 8t$. Find its speed at $t = 3$.

a. 6

b. 10 ✓

c. 8

d. 12

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 8$. At $t = 3$: $\frac{dx}{dt} = 6$. Speed $= \sqrt{6^2 + 8^2} = \sqrt{100} = 10$.

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41 [Challenging] • Target: First and Second Derivatives Parametrically • Cross-Lesson: 3.02 Vector-Valued Functions

A curve is defined by $x(t) = t^2 + 1$ and $y(t) = t^3 - 3t$. Find $\frac{dy}{dx}$ at $t = 2$.

a. $\frac{9}{4}$ ✓

b. $\frac{3}{2}$

c. $\frac{4}{9}$

d. 9

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 3t^2 - 3$. $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2-3}{2t}$. At $t = 2$: $\frac{12-3}{4} = \frac{9}{4}$.

42 [Challenging] • Target: First and Second Derivatives Parametrically • Cross-Lesson: 4.01 Polar Curves and Derivatives

For $x(t) = t^2$ and $y(t) = t^3$, find $\frac{d^2y}{dx^2}$.

a. $\frac{3}{4t}$ ✓

b. $\frac{3}{2}$

c. $\frac{6t}{2}$

d. $\frac{3}{4t^2}$

$\frac{dy}{dx} = \frac{3t^2}{2t} = \frac{3}{2}t$. Then $\frac{d}{dt}[\frac{3}{2}t] = \frac{3}{2}$. Dividing by $\frac{dx}{dt} = 2t$ gives $\frac{3/2}{2t} = \frac{3}{4t}$.

43 [Challenging] • Target: Horizontal and Vertical Tangents • Cross-Lesson: 4.02 Area in Polar Coordinates

Find the equation of the tangent line at $t = 1$ to the curve $x(t) = e^{2t}, y(t) = 3t^2 - 1$.

a. $y - 2 = \frac{3}{e^2}(x - e^2)$ ✓

b. $y - 2 = 3(x - e^2)$

c. $y = \frac{6}{e^2}x$

d. $y - 1 = \frac{3}{e^2}(x - e^2)$

At $t = 1$: $x = e^2, y = 2$. $\frac{dx}{dt} = 2e^{2t} = 2e^2$, $\frac{dy}{dt} = 6t = 6$. Slope $= \frac{6}{2e^2} = \frac{3}{e^2}$. Equation: $y - 2 = \frac{3}{e^2}(x - e^2)$.

44 [Challenging] • Target: Arc Length of Parametric Curves • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Find the arc length of the curve $x = 3\cos t, y = 3\sin t$ from $t = 0$ to $t = \pi$.

a. 3π ✓

b. 6π

c. 9π

d. $\frac{3\pi}{2}$

$x' = -3\sin t, y' = 3\cos t$. $\sqrt{(-3\sin t)^2 + (3\cos t)^2} = \sqrt{9} = 3$. Length $= \int_0^{\pi} 3dt = 3\pi$ (semicircle of radius 3).

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45 [Challenging] • Target: Speed and Total Distance Traveled • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

A particle moves in the xy-plane with velocity vector $\mathbf{v}(t) = \langle 4t, 3 \rangle$. What is its speed at $t = 1$?

a. 5 ✓

b. 7

c. 4

d. $\sqrt{7}$

Speed is the magnitude of the velocity vector: $||\mathbf{v}(1)|| = \sqrt{(4(1))^2 + 3^2} = \sqrt{16 + 9} = 5$.

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